James, Oluwafemi Gabriel - 000937409

Emergency Shelters Daily Occupancy And Capacity in Alberta, Canada.

Introduction (Business Understanding)

This dataset reveals patterns in shelter usage across Alberta from 2013 to 2024. With a growing need for shelter services, we aim to uncover trends, predict future demands, and evaluate capacity readiness for 2025 and 2026.

Beyond forecasting future trends, understanding which cities or organizations are most at risk during critical demand periods can guide resources allocation and emergency planning.

Business Questions

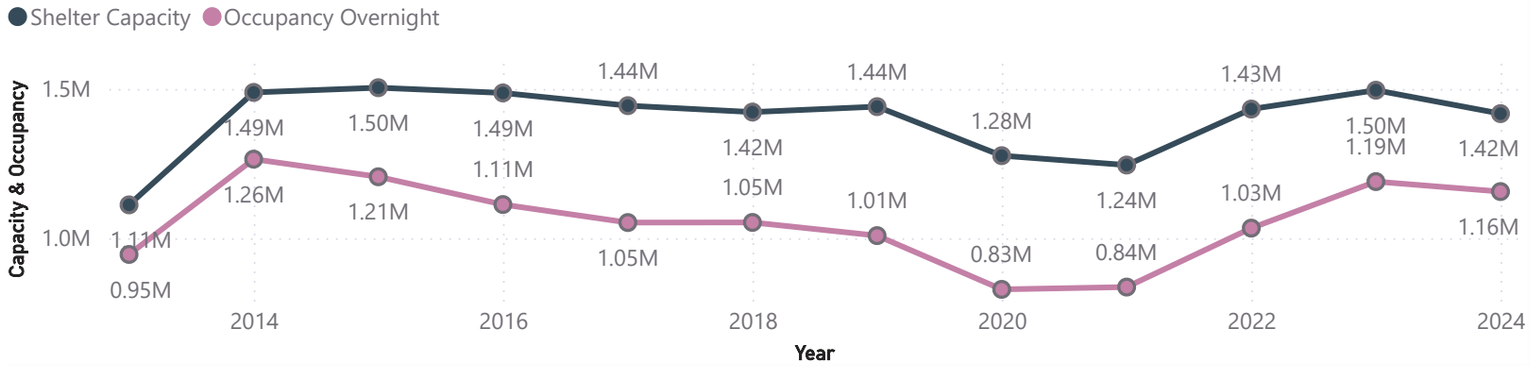
1. Will a 20% yearly increase in occupancy through 2026 exceed available Capacity?
2. Which cities and organizations face the highest & lowest risk of shelter overcapacity during peak demand periods?

Business Intelligence Reporting - DATA 460

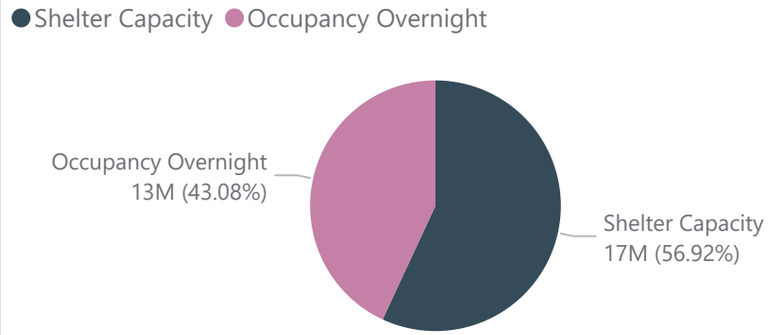
Reference: This collection of datasets are provided from 2013-2024, on the number of individuals accessing Adult Shelters funded by Seniors, Community and Social Services (SCSS). Information contained in the data set includes: Date, Shelter Name, Total Approved Beds, Number of Beds funded by SCSS, Number of Beds funded by SCSS for Winter Emergency Response, Daily Occupancy.
Link: https://open.alberta.ca/opendata/funded-emergency-shelters-daily-occupancy-ab/resource/b7080b66-25ea-4c30-ac47-02b64353637f

Data Understanding of Emergency Shelters Daily Occupancy And Capacity in Alberta, Canada.

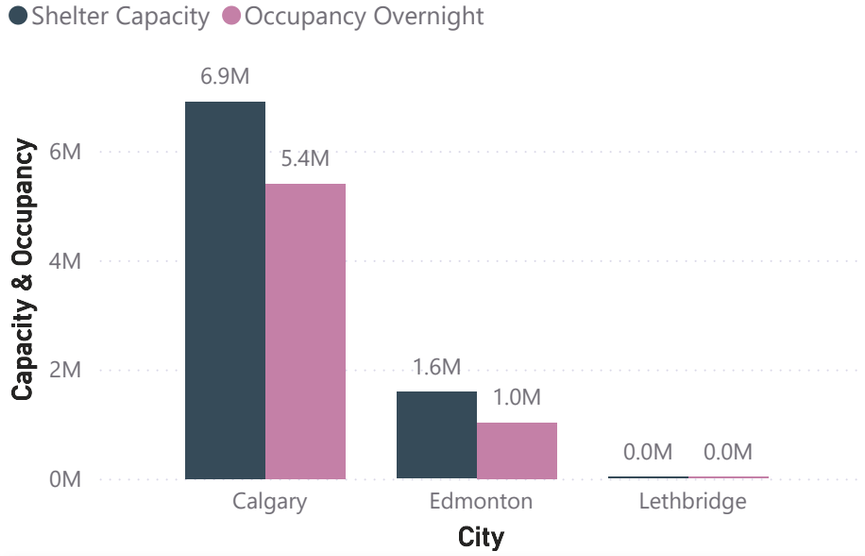
Homeless Shelter Capacity & Occupancy Overnight By Year in Alberta (2013 - 2024)



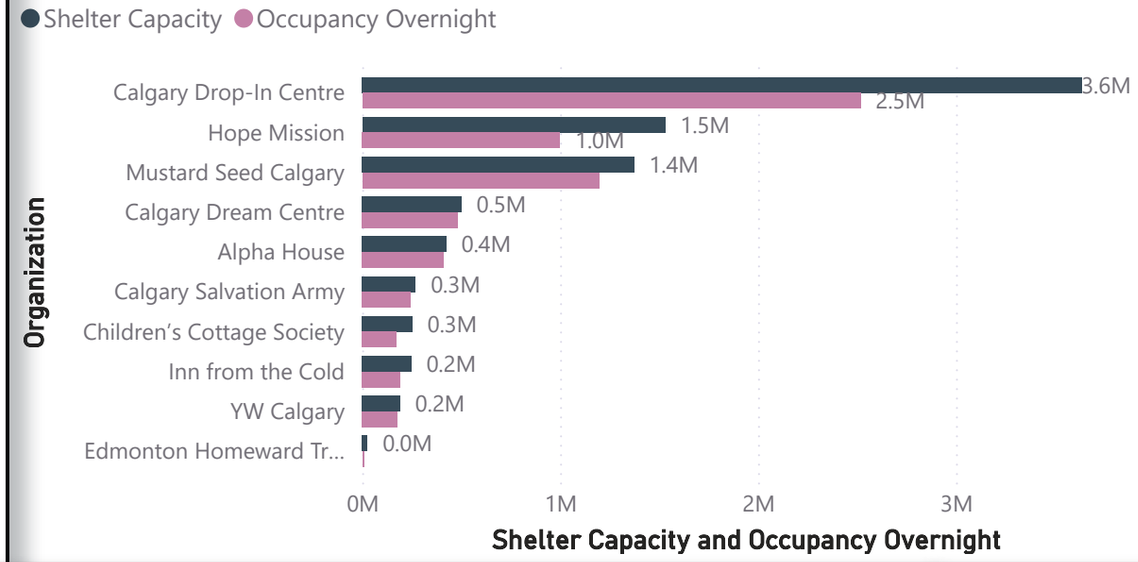
Shelter Capacity & Occupancy Overnight in Alberta



Shelter Capacity & Occupancy Overnight by City



Top 10 Shelter Capacity & Occupancy Overnight by Organization



Organizational Capacity and Utilization:

- **Top Organizations:**
  - **Calgary Drop-In Centre:** Has the highest capacity (2.52M) but a notable gap in overnight occupancy (1.64M).
  - **Hope Mission:** Follows with significant occupancy (1.53M), almost matching its capacity.
  - **Mustard Seed Calgary:** Shows high occupancy (1.37M) relative to its capacity.

Other Organizations: Smaller shelters like Edmonton Homeward Trust and YW

Trend Over Time (2013–2024):

- **Shelter Capacity:** Shows a steady increase from 1.14M in 2013 to a peak of 1.50M in 2022, followed by a slight decline to 1.42M in 2024.
- **Occupancy Overnight:** Peaks at 1.26M in 2014 but has a gradual decline, hitting its lowest point at 0.83M in 2020, before stabilizing around 1.16M by 2024. This suggests a gap between capacity and utilization has widened over time.

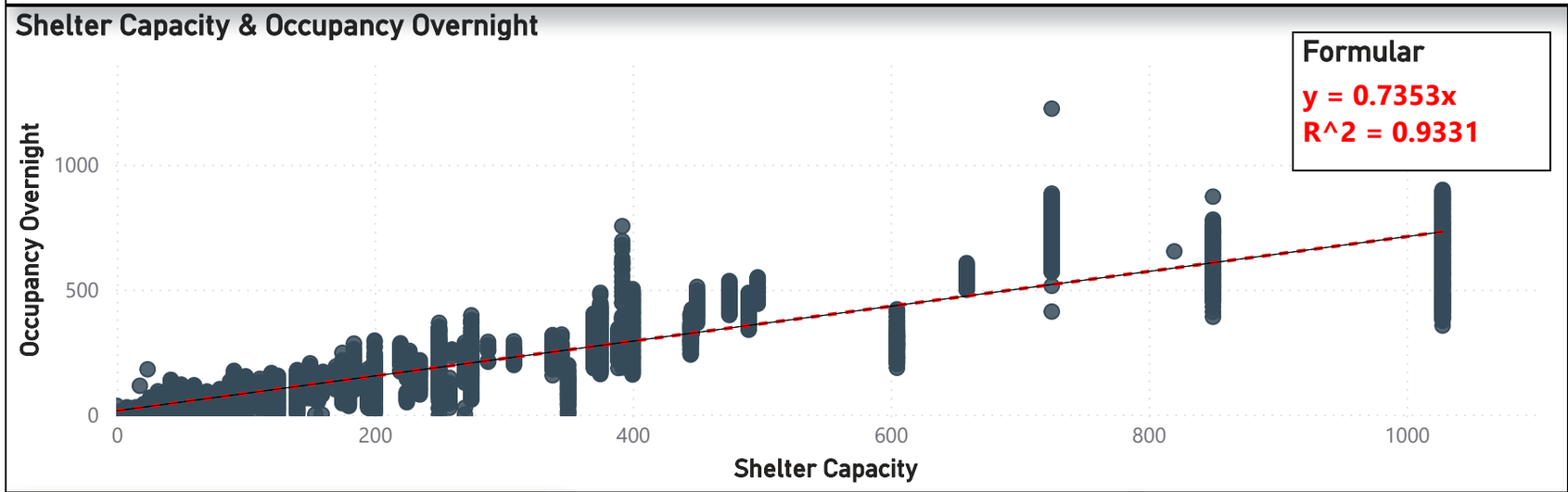
City-Level Analysis:

- **Calgary:** Dominates both shelter capacity (6M) and occupancy overnight (about 4M), reflecting a higher concentration of homelessness services.
- **Edmonton:** Has a smaller capacity and occupancy than Calgary, with a visible gap.
- **Lethbridge:** Has significantly lower numbers, indicating fewer shelters or lower demand.

Overall Capacity vs. Occupancy:

- **Aggregate Numbers:** Shelter capacity (17M) outpaces occupancy overnight (13M), with occupancy representing 43% of total capacity. This reflects underutilization and could point to accessibility issues or changing demands.

Diagnostic Analysis - Relationship Between Shelter Capacity & Occupancy in Alberta, Canada.



Diagnostic Findings

1. **High Correlation**:
   - R^2 = 0.9331 suggest a very strong relationship between shelter capacity and occupancy, suitable for predictive modeling.
   - The slope 0.7353 indicates that for every 1-unit increase in Capacity, the expected value of Overnight increases by approximately 0.7353 units.
   - The trend suggests that occupancy scales with capacity, but slightly underutilizes available shelter space.

Prediction Setup:

- The diagnostic insights support forecasting for future trends, especially for capacity utilization under increasing occupancy scenarios.

Average Shelter Capacity	Peak Capacity	Average Occupancy Overnight	Peak Occupancy Overnight
60.45	1028	45.59	1224

Answering Business Questions

1. **Will a 20% yearly increase in occupancy through 2026 exceed available capacity?**
   - Use DAX formula to **Forecast Occupancy 2026** to predict occupancy values for 2026.
   - Compare predicted occupancy to capacity using **Capacity Utilization 2026 (%).**
   - **Steps**:
     - Identify shelters or cities where **Capacity Utilization 2026 (%)** exceeds 100% (overcapacity risk).
     - Summarize findings for Alberta.
2. **Which cities and organizations face the highest risk of shelter overcapacity during peak demand periods?**
   - Leverage **Peak Occupancy** and **Risk Factor (%)** measures.
   - Highlight cities/organizations where **Risk Factor (%) > 100%**.
   - **Steps**:
     - Sort shelters and cities by **Risk Factor (%).**
     - Identify “High Risk” shelters flagged by DAX **High Risk** column.

Answering Business Questions

1. **Will a 20% yearly increase in occupancy through 2026 exceed available capacity?**

Yes; This increase in occupancy will exceed capacity in 2026 in

1. Homelessness Society of the Bow Valley, Bow Valley by 17.21%
2. Mustard Seed, Calgary by 15.20%
3. Niginan Housing Ventures, Edmonton by 0.75%

Organization	city	2025 Capacity Utilization (%)	2025 Forecast Occupancy	2026 Capacity Utilization (%)	2026 Forecast Occupancy
Homelessness Society of the Bow Valley	Bow Valley	97.68	656.40	117.21	787.68
Mustard Seed	Calgary	96.00	14,668.80	115.20	17,602.56
Niginan Housing Ventures	Edmonton	83.96	20,234.40	100.75	24,281.28
Salvation Army	Edmonton	73.25	51,591.60	87.90	61,909.92
Enoch Cree Nation	Edmonton	72.21	21,447.60	86.66	25,737.12
Mustard Seed	Edmonton	65.28	32,056.80	78.34	38,468.16
Elizabeth Fry Society of Northern Alberta	Edmonton	64.45	12,374.40	77.34	14,849.28
Blood Tribe Department of Health	Lethbridge	55.74	47,564.40	66.89	57,077.28
Town of Slave Lake	Slave Lake	50.17	2,767.20	60.20	3,320.64
Hope Mission	Wetaskiwin	40.60	14,913.60	48.72	17,896.32
Mustard Seed Medicine Hat	Medicine Hat	39.38	11,163.60	47.25	13,396.32
Sagitawa Friendship Centre	Peace River	39.07	7,935.60	46.89	9,522.72
John Howard Society	Cold Lake	30.70	3,831.60	36.84	4,597.92
Safe Harbour Society	Red Deer	19.01	61,582.80	22.82	73,899.36
Town of Drayton Valley	Drayton Valley	17.83	4,149.60	21.40	4,979.52
Mustard Seed Red Deer	Red Deer	17.76	16,156.80	21.31	19,388.16
Mustard Seed Edmonton	Edmonton	17.30	15,033.60	20.76	18,040.32
Leduc Hub	Leduc	16.46	1,326.00	19.76	1,591.20
Lakeland Out of the Elements	Lac La Biche	12.59	2,556.00	15.10	3,067.20
Hope Mission	Edmonton	9.66	289,621.20	11.60	347,545.44
Calgary Dream Centre	Calgary	9.56	51,079.20	11.47	61,295.04
Total		**8.28**	**1,387,165.20**	**9.93**	**1,664,598.24**

Answering Business Questions

2. Which cities and organizations face the highest risk of shelter overcapacity during peak demand periods?

City	Organization	High Risk	Risk Factor (%)	Peak Occupancy
Edmonton	Elizabeth House	High Risk	754.17	181
Grande Prairie	Elders Caring Centre	High Risk	319.44	115
Edmonton	Mustard Seed	High Risk	102.22	46
Calgary	The Alex	High Risk	101.43	71
Edmonton	Operation Friendship	High Risk	43.04	68
Cold Lake	John Howard Society	High Risk	36.67	22
Cold Lake	Dr. Margaret Savage Crisis Centre	High Risk	22.22	16
Edmonton	Edwardson Place	High Risk	17.36	25
Grande Prairie	City of Grande Prairie	High Risk	9.60	24
Lethbridge	Lethbridge Shelter and Resource Centre	High Risk	8.05	143
Edmonton	Boyle McCauley	High Risk	7.50	77
Drayton Valley	Town of Drayton Valley	High Risk	6.67	16
Red Deer	City of Red Deer	High Risk	5.54	62
Calgary	Children's Cottage Society	High Risk	4.11	79
Lac La Biche	Lakeland Out of the Elements	High Risk	3.61	15
Edmonton	Holy Trinity Lutheran Church	High Risk	3.41	53
Bow Valley	Homelessness Society of the Bow Valley	High Risk	3.41	9
Slave Lake	Slave Lake Homeless Coalition	High Risk	3.25	24
Calgary	Calgary YWCA	High Risk	2.98	64
Edmonton	Niginan Housing Ventures	High Risk	2.89	55
Wetaskiwin	Hope Mission	High Risk	2.51	66
Peace River	Sagitawa Friendship Centre	High Risk	2.43	29
Edson	Reflections Society for Empowering People to Succeed	Low Risk	2.33	5
Wetaskiwin	Mustard Seed Wetaskiwin	High Risk	2.20	33
Calgary	YW Calgary	High Risk	2.11	99
Lethbridge	Lethbridge Stabilization Centre	High Risk	1.96	73
Edmonton	Mustard Seed Edmonton	High Risk	1.95	192
Total			**0.01**	**1224**

Leveraging Peak Occupancy and Risk Factor (%) measures.

The cities and organizations that face the highest risk of shelter overcapacity during peak demand periods where Risk Factor (%) > 100% are;

1. **Elizabeth House, Edmonton** with **754.17%** Risk Factor.
2. **Elders Caring Centre, Grande Prairie** with **319.44%** Risk Factor.
3. **Mustard Seed, Edmonton** with **2.22%** above 100% Risk Factor.
4. **The Alex, Calgary** with **1.43%** above 100% Risk Factor.

The cities and organizations that face the lowest risk of shelter overcapacity during peak demand periods are;

1. **Wapiti Community Support Association, Grande Prairie** with **0.01%** Risk Factor
2. **Operation Friendship, Edmonton** with 0.02% Risk Factor.

Conclusion

Based on the diagnostic and predictive analysis, Alberta faces significant challenges in emergency shelter management if occupancy trends continue to rise. By 2026, several shelters, including Homelessness Society of the **Bow Valley, Mustard Seed Calgary, and Niginan Housing Ventures,** are projected to exceed capacity, posing risks to vulnerable populations.

Additionally, shelters such as Elizabeth **House, Edmonton, and Elders Caring Centre, Grande Prairie,** face extremely high risk during peak periods, requiring immediate attention.

Recommendations

Strategic Resource Allocation:

1. Prioritize funding and expansion for shelters identified as high risk, focusing on Edmonton, Grande Prairie, and Calgary.
2. Implement policies to address capacity gaps, such as increasing capacity or reducing demand through housing initiatives.

Early Warning and Monitoring:

1. Use predictive models to monitor occupancy trends quarterly and flag high-risk periods or shelters.
2. Deploy emergency response teams during forecasted peak demand periods.